

Yuxing Wang (王昱兴)

Designing Machines That Design Machines

✉ wyx20@mails.tsinghua.edu.cn <https://yuxing-wang-thu.github.io/>

RESEARCH INTERESTS

My research focuses on **embodied co-design**, **foundation models**, and **agentic systems**. I am particularly interested in building autonomous systems that can co-design the morphology and control of embodied agents, enabling them to be both physically viable and behaviorally adaptive in complex, unconventional environments. My previous work spans evolutionary reinforcement learning for robot control optimization and morphology-control co-optimization for modular soft robots. I have also worked on applications in Game AI and LLM-assisted decision-making systems.

EDUCATION AND APPOINTMENTS

Tsinghua University <i>PhD Student in Control Science and Engineering</i> • Advisor: Prof. Xueqian Wang and Prof. Zhiheng Li • GPA: 4.00/4.00; Rank: 1/352	Aug. 2025 – Present
Tsinghua University <i>Research Assistant in Tsinghua SIGS</i>	Jul. 2023 – Jul. 2025
Tsinghua University <i>M.Eng. in Electronic and Information Engineering</i> • Advisor: Prof. Xueqian Wang • GPA: 3.89/4.00; Rank: 2/45	Aug. 2020 – Jun. 2023
Southwest Minzu University <i>B.Eng. in Communication Engineering, with Highest Honors</i> • GPA: 3.86/4.00; Rank: 1/145	Sep. 2016 – Jun. 2020

PUBLICATIONS

First-Author and Co-First-Author Publications

1. X. Li, **Y. Wang**^{*}, X. Wang, and H. Liu, “Enhancing Quality-Diversity with Adaptive Heterogeneous Operator Allocation,” *Parallel Problem Solving From Nature (PPSN)*, [CCF B], Mar. 2026.
2. C. Yu, **Y. Wang**^{*}, Z. Yang, X. Li, and X. Wang, “Surrogate-Assisted Evolutionary Multi-Agent Reinforcement Learning with Adaptive Fitness Evaluation,” *Genetic and Evolutionary Computation Conference (GECCO)*, [CCF C], Mar. 2026.
3. **Y. Wang**, S. Huang, J. Li, C. Yu, X. Li, Y. Chang, X. Wang, and B. Liang, “Morphology and Behavior Co-Optimization of Modular Satellites for Attitude Control,” *75th International Astronautical Congress (IAC)*, [Oral Presentation], Sep. 2024.
4. **Y. Wang**, S. Wu, T. Zhang, Y. Chang, H. Fu, Q. Fu, and X. Wang, “PreCo: Enhancing Generalization in Co-Design of Modular Soft Robots via Brain-Body Pre-Training,” *Conference on Robot Learning (CoRL)*, [Oral Presentation; Top 6.6%], Nov. 2023.
5. **Y. Wang**, S. Wu, H. Fu, Q. Fu, T. Zhang, Y. Chang, and X. Wang, “Curriculum-Based Co-Design of Morphology and Control of Voxel-Based Soft Robots,” *International Conference on Learning Representations (ICLR)*, [CCF A], Jan. 2023.
6. **Y. Wang**, T. Zhang, Y. Chang, B. Liang, X. Wang, and B. Yuan, “A Surrogate-Assisted Controller for Expensive Evolutionary Reinforcement Learning,” *Information Sciences*, [JCR Q1], Dec. 2021.
7. **Y. Wang** and B. Yuan, “From Big-Data-Based Price Discrimination to Privacy Leakage: Ethical Analysis and Reflections on Privacy Issues from the Perspective of Hardware and Software,” *Science Economy Society*, [Chinese Journal], Oct. 2021.

8. **Y. Wang** and Y. Jiang, “A Weighted Minimum Distance Classifier Based on Relative Offset,” *IEEE 4th International Conference on Cloud Computing and Big Data Analysis (ICCCBDA)*, [EI], Apr. 2019.

Selected Preprints and Manuscripts

9. **Y. Wang**, Z. Chen, T. Zhang, L. Wang, and X. Wang, “Embodied Co-Design for Rapidly Evolving Agents: Taxonomy, Frontiers, and Challenges,” *Preprint*, [Submitted to IEEE TEVC], Dec. 2025.

Other Publications

10. X. Li, X. Xu, **Y. Wang**, X. Wang, and H. Liu, “Multi-Objective Fault-Tolerant Trajectory Planning for Space Manipulators: A Quality-Diversity Based Approach and Validation,” *76th International Astronautical Congress (IAC)*, [Oral Presentation], Aug. 2025.
11. C. Yu, S. Jiang, **Y. Wang**, and X. Wang, “Autonomous Trajectory Planning of Free-Floating Space Manipulators: A Diversity-Enhanced Evolutionary Reinforcement Learning Approach,” *76th International Astronautical Congress (IAC)*, [Poster Presentation], Aug. 2025.
12. T. Zhang, Z. Lin, **Y. Wang**, D. Ye, Q. Fu, X. Wang, B. Liang, and X. Li, “Dynamics-Adaptive Continual Reinforcement Learning via Progressive Contextualization,” *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, [JCR Q1], Aug. 2022.

RESEARCH EXPERIENCE

Quality–Similarity Diversity Optimization for User-Preferred Game AI May 2022 – Apr. 2023

Tencent AIPD; Supported by the Tencent Rhino-Bird Research Elite Program

- Developed a general-purpose framework for quality–similarity diversity optimization, enabling the generation of task-specific behavioral diversity for game AI based on user-specified behavior descriptors.
- Applied the framework to multiple game environments, including Atari 2600 and Auto Chess.

Human versus Machine Intelligence Jan. 2020 – Dec. 2021

Tsinghua University; Supported by the National Natural Science Foundation of China

- Constructed distributed Deep Q-Networks and Evolution Strategies for training reinforcement learning agents.
- Designed and implemented the overall training framework for game-playing agents.
- Developed agents that achieved competitive performance against expert-level human players.

Wearable Hand Rehabilitation Robot Aug. 2018 – Dec. 2019

Southwest Minzu University; National Undergraduate Innovation and Entrepreneurship Training Program

- Designed and fabricated a wearable hand rehabilitation robot for assisting patients with finger flexion and extension exercises.
- Manufactured mechanical components using 3D printing, debugged the electronic circuit system, and developed a web server for rehabilitation data collection.

International Underwater Robot Competition, 2D Simulation League Jun. 2017 – Jun. 2018

Southwest Minzu University–Peking University Joint Underwater Robot Lab

- Developed control strategies for 2D simulated robotic fish using C#.
- Led the team to win First Prize in the International Underwater Robot Competition in both 2017 and 2018.

ACADEMIC SERVICE

Membership IEEE Member; CIE Member

Journal Reviewer IEEE TNNLS; Information Sciences; IEEE Access

Conference Reviewer NeurIPS; ICLR; ICML; RSS; CoRL; ACM MM

AWARDS AND SCHOLARSHIPS

- National Scholarship 2017–2019
- First Prize, International Underwater Robot Competition Jul. 2017

- Honorable Mention, Mathematical Contest in Modeling Apr. 2018
- First Prize, Central and Southwestern China Division, “TI Cup” National Undergraduate IoT Design Contest Aug. 2018
- Bronze Award, National College Student Curricular Academic Science and Technology Works Competition Mar. 2019
- Outstanding Graduate of Sichuan Province Apr. 2020
- *People’s Daily* Representative List of National Scholarship Winners for Undergraduate Students Apr. 2020
- Tencent Rhino-Bird Research Elite Program May 2022

TEACHING, SKILLS, AND ACTIVITIES

Teaching	Teaching Assistant for <i>Aircraft flight control system</i> , <i>Data Mining</i> and <i>Ethics of Artificial Intelligence</i> , Tsinghua University.
Technical Skills	Python, PyTorch, TensorFlow, LaTeX, Linux, C++...
Certification	Teacher Qualification Certificate.
Interests	J-pop music, traveling.
Talks	Brain-Body Co-Design for Embodied Agents: An Introduction and Recent Advances, 05/2024, The Chinese University of Hong Kong, Shenzhen, China. [PPT]
News Report	我设计我自己! 清华最新研究: 未来机器人不用人设计, AI 直接捏出最优形态, 02/2026, DeepTech.